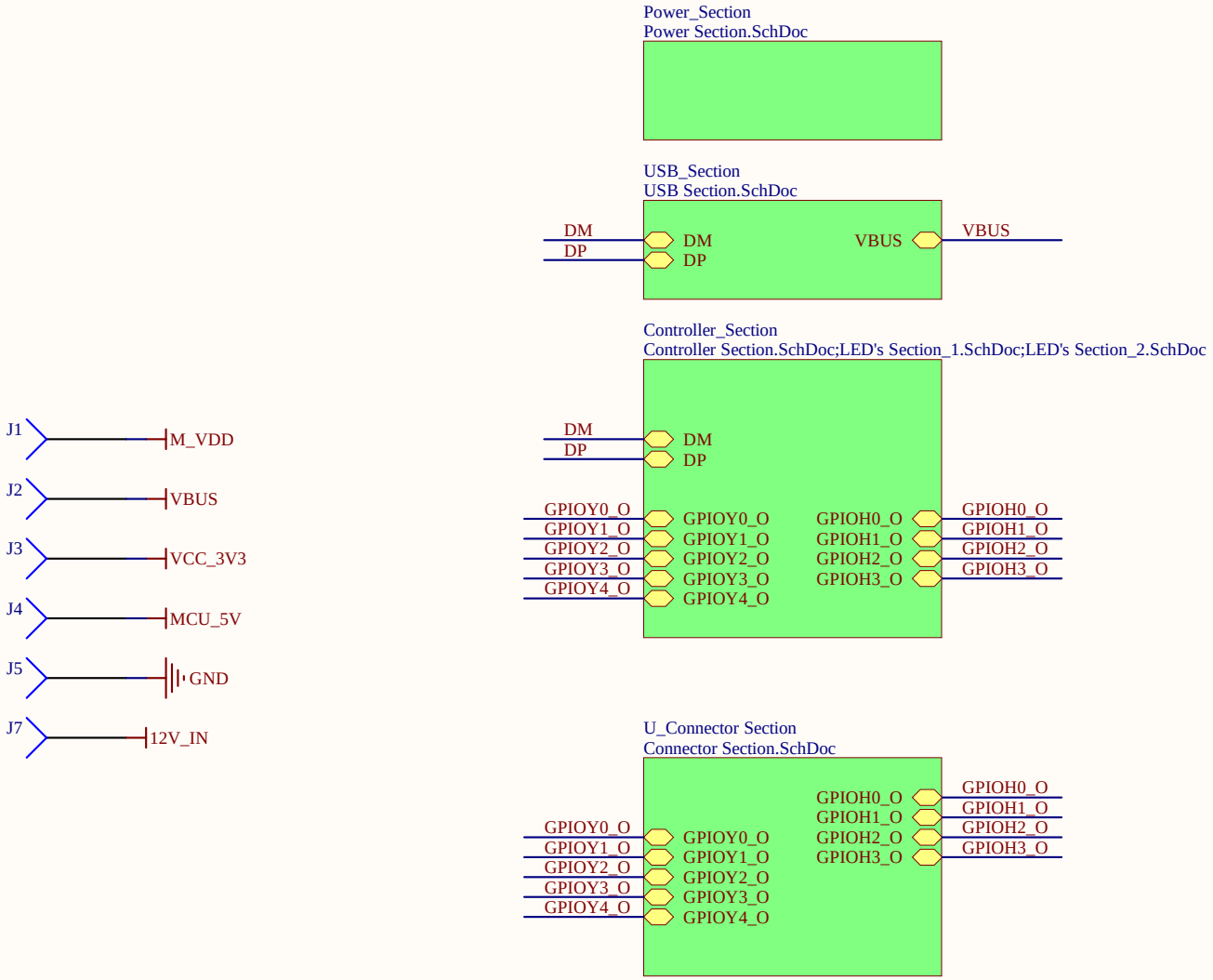
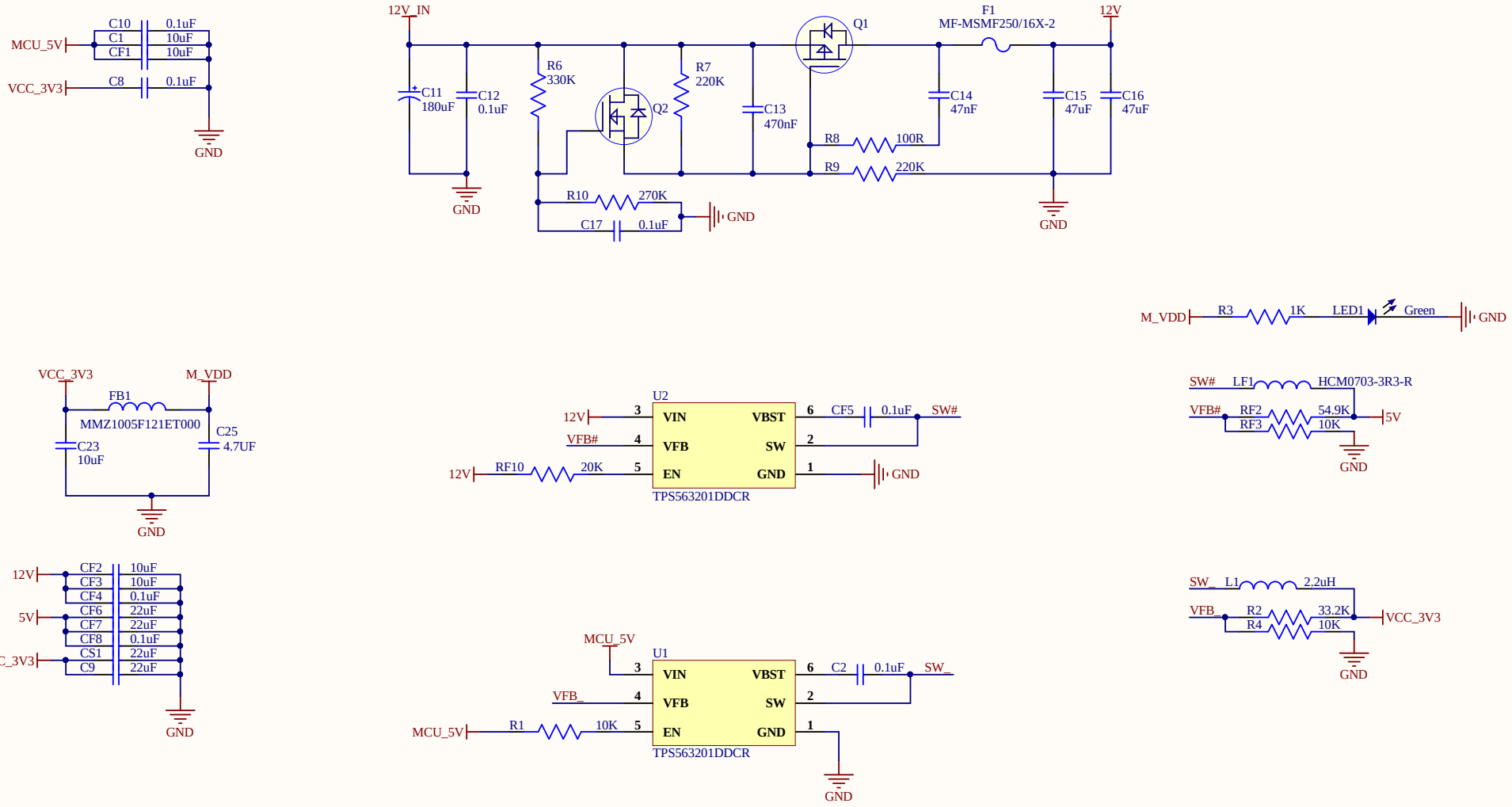
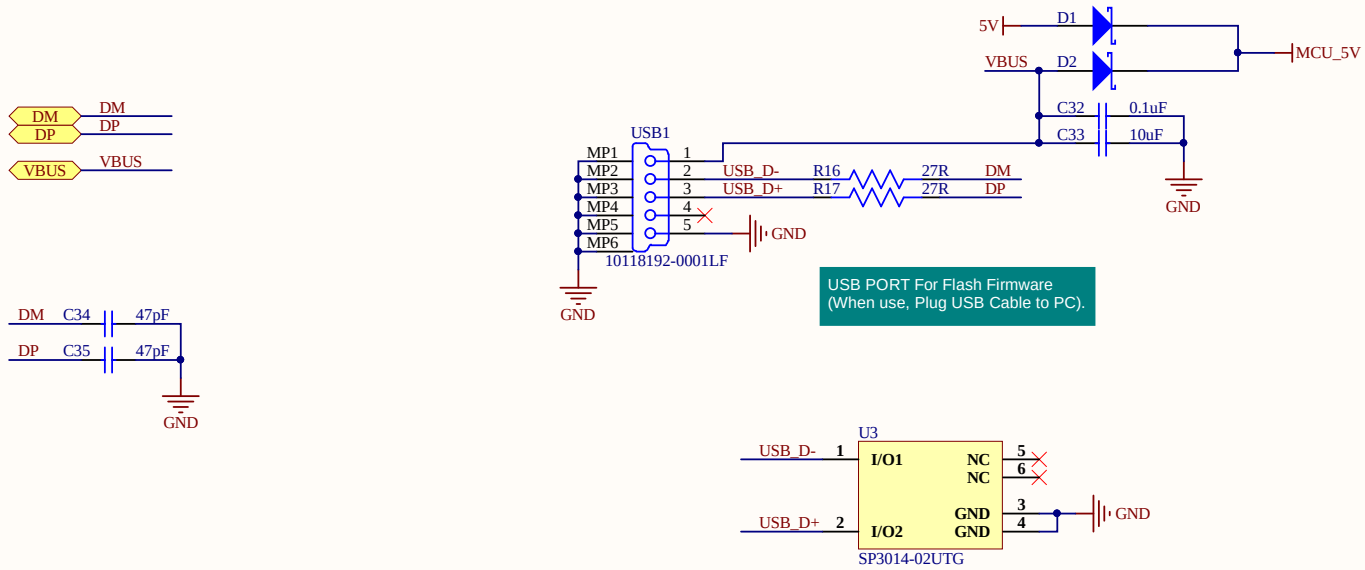
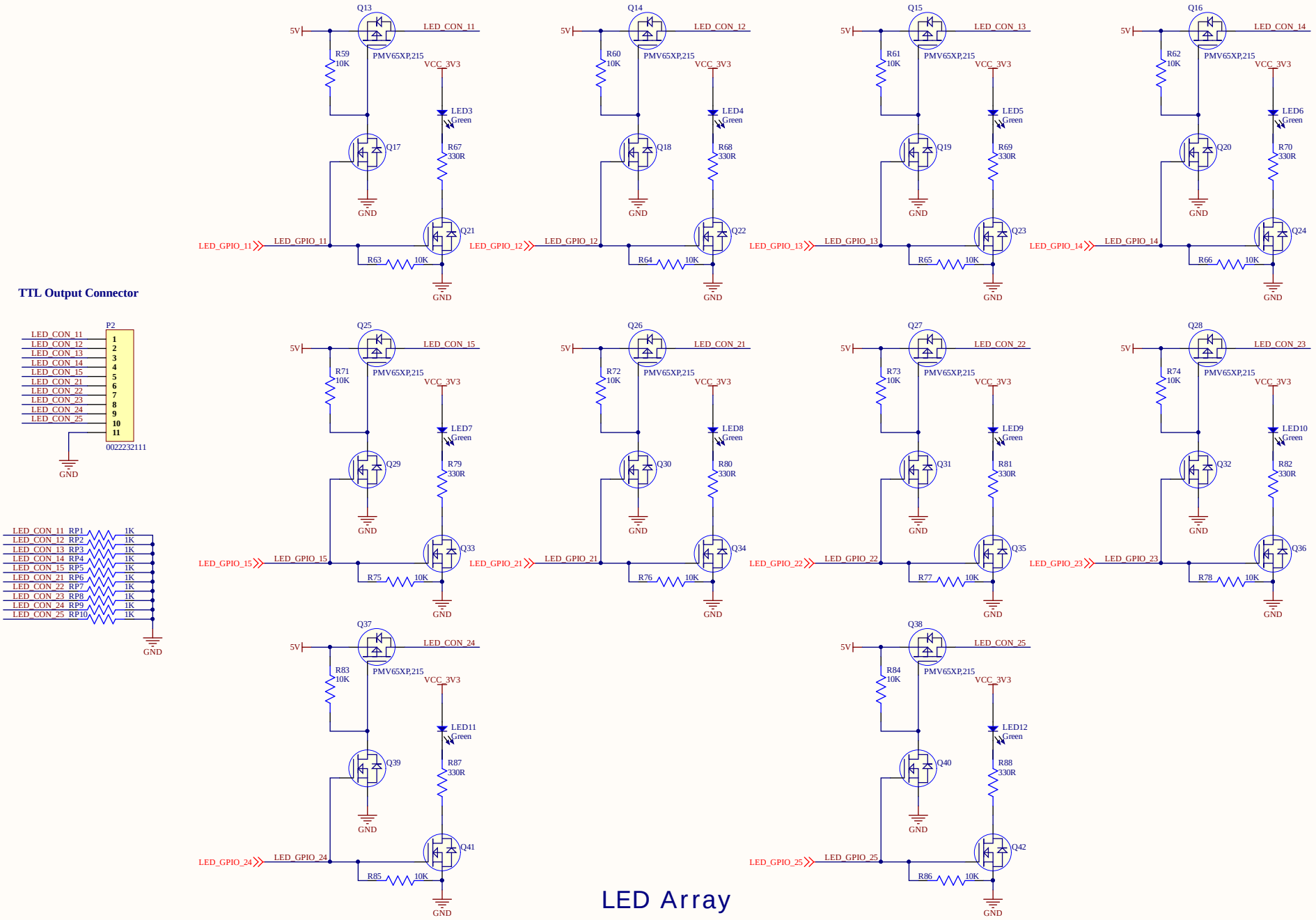

Power_Section
Power Section.SchDoc
USB_Section
USB Section.SchDoc
DM
DP
DM
DP
VBUS
VBUS
Controller_Section
Controller Section.SchDoc;LED's Section_1.SchDoc;LED's Section_2.SchDoc
DM
DP
DM
DP
GPIOY0_O
GPIOY1_O
GPIOY2_O
GPIOY3_O
GPIOY4_O
GPIOY0_O
GPIOY1_O
GPIOY2_O
GPIOY3_O
GPIOY4_O
GPIOH0_O
GPIOH1_O
GPIOH2_O
GPIOH3_O
GPIOH0_O
GPIOH1_O
GPIOH2_O
GPIOH3_O
U_Connector Section
Connector Section.SchDoc
GPIOY0_O
GPIOY1_O
GPIOY2_O
GPIOY3_O
GPIOY4_O
GPIOY0_O
GPIOY1_O
GPIOY2_O
GPIOY3_O
GPIOY4_O
GPIOH0_O
GPIOH1_O
GPIOH2_O
GPIOH3_O
GPIOH0_O
GPIOH1_O
GPIOH2_O
GPIOH3_O
J1
M_VDD
J2
VBUS
J3
VCC_3V3
J4
MCU_5V
J5
GND
J7
12V_IN

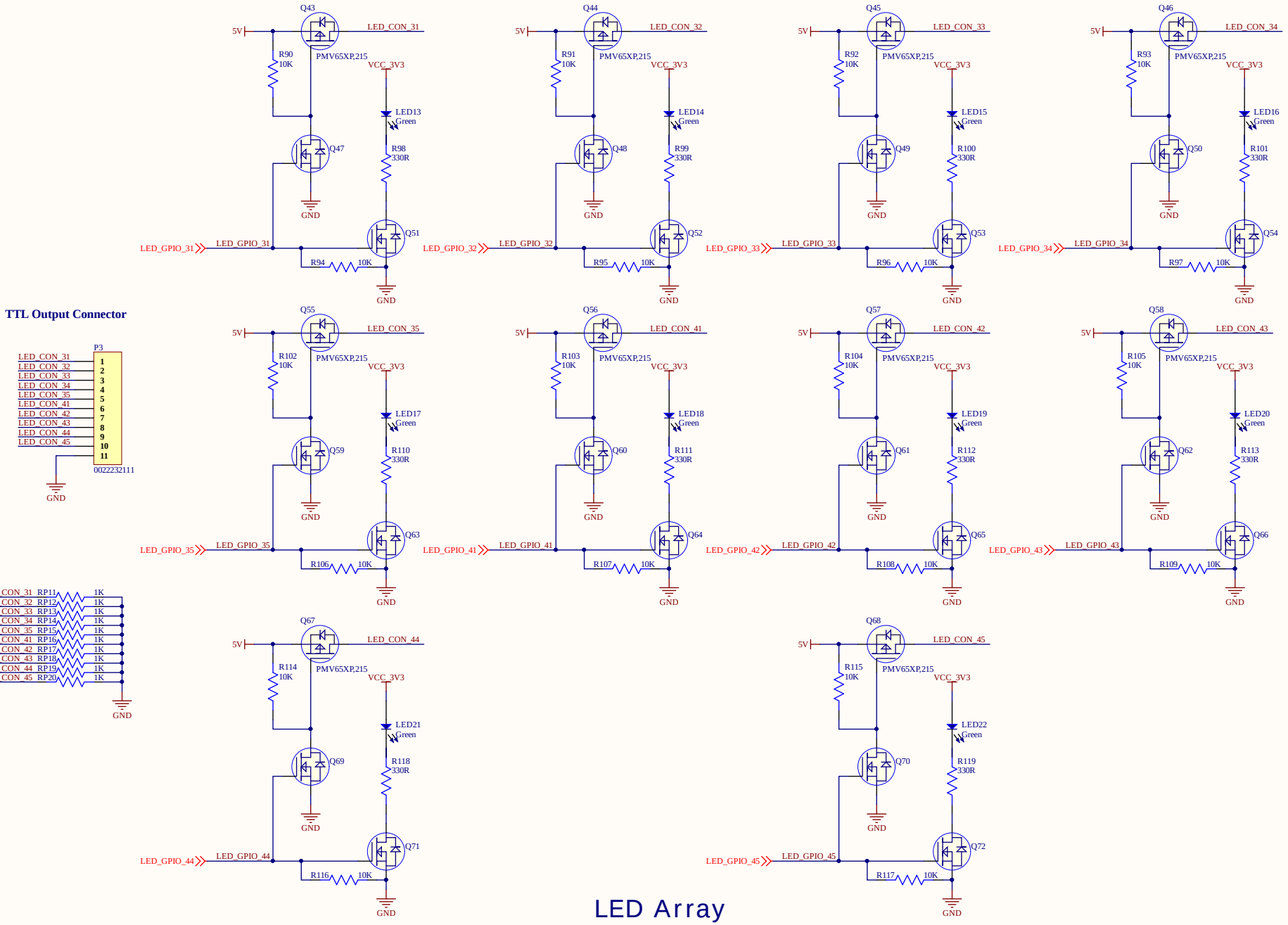


Power Section

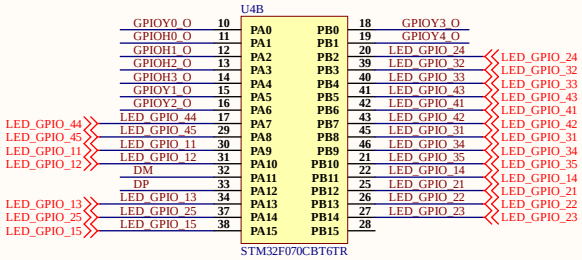

DM
DP
VBUS
DM
DP
VBUS
DM C34 47pF
DP C35 47pF
GND
USB1
MP1
MP2
MP3
MP4
MP5
MP6
1
2
3
4
5
USB_D-
USB_D+
R16
R17
27R
27R
DM
DP
GND
10118192-0001LF
GND
5V
VBUS
D1
D2
MCU_5V
C32 0.1uF
C33 10uF
GND
USB PORT For Flash Firmware
(When use, Plug USB Cable to PC).
U3
USB_D- 1
USB_D+ 2
I/O1
I/O2
NC
NC
GND
GND
5
6
3
4
GND
SP3014-02UTG

USB Section

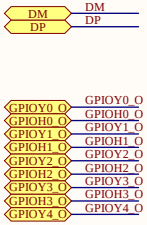

TTL Output Connector
P2
LED_CON_11
LED_CON_12
LED_CON_13
LED_CON_14
LED_CON_15
LED_CON_21
LED_CON_22
LED_CON_23
LED_CON_24
LED_CON_25
1
2
3
4
5
6
7
8
9
10
11
0022232111
GND
LED_CON_11 RP1 1K
LED_CON_12 RP2 1K
LED_CON_13 RP3 1K
LED_CON_14 RP4 1K
LED_CON_15 RP5 1K
LED_CON_21 RP6 1K
LED_CON_22 RP7 1K
LED_CON_23 RP8 1K
LED_CON_24 RP9 1K
LED_CON_25 RP10 1K
Q13
Q14
Q15
Q16
Q17
Q18
Q19
Q20
Q21
Q22
Q23
Q24
Q25
Q26
Q27
Q28
Q29
Q30
Q31
Q32
Q33
Q34
Q35
Q36
Q37
Q38
Q39
Q40
Q41
Q42
PMV65XP,215
5V
VCC_3V3
LED_CON_11
LED_CON_12
LED_CON_13
LED_CON_14
LED_CON_15
LED_CON_21
LED_CON_22
LED_CON_23
LED_CON_24
LED_CON_25
R59 10K
R60 10K
R61 10K
R62 10K
R63 10K
R64 10K
R65 10K
R66 10K
R67 330R
R68 330R
R69 330R
R70 330R
R71 10K
R72 10K
R73 10K
R74 10K
R75 10K
R76 10K
R77 10K
R78 10K
R79 330R
R80 330R
R81 330R
R82 330R
R83 10K
R84 10K
R85 10K
R86 10K
R87 330R
R88 330R
LED3 Green
LED4 Green
LED5 Green
LED6 Green
LED7 Green
LED8 Green
LED9 Green
LED10 Green
LED11 Green
LED12 Green
LED_GPIO_11
LED_GPIO_12
LED_GPIO_13
LED_GPIO_14
LED_GPIO_15
LED_GPIO_21
LED_GPIO_22
LED_GPIO_23
LED_GPIO_24
LED_GPIO_25
GND
LED Array

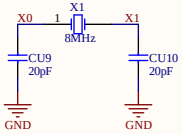

TTL Output Connector
P3
LED_CON_31
LED_CON_32
LED_CON_33
LED_CON_34
LED_CON_35
LED_CON_41
LED_CON_42
LED_CON_43
LED_CON_44
LED_CON_45
0022232111
GND
LED_CON_31 RP11 1K
LED_CON_32 RP12 1K
LED_CON_33 RP13 1K
LED_CON_34 RP14 1K
LED_CON_35 RP15 1K
LED_CON_41 RP16 1K
LED_CON_42 RP17 1K
LED_CON_43 RP18 1K
LED_CON_44 RP19 1K
LED_CON_45 RP20 1K
Q43
Q44
Q45
Q46
Q55
Q56
Q57
Q58
Q67
Q68
PMV65XP,215
5V
VCC_3V3
R90 10K
R91 10K
R92 10K
R93 10K
R102 10K
R103 10K
R104 10K
R105 10K
R114 10K
R115 10K
LED13 Green
LED14 Green
LED15 Green
LED16 Green
LED17 Green
LED18 Green
LED19 Green
LED20 Green
LED21 Green
LED22 Green
R98 330R
R99 330R
R100 330R
R101 330R
R110 330R
R111 330R
R112 330R
R113 330R
R118 330R
R119 330R
Q47
Q48
Q49
Q50
Q59
Q60
Q61
Q62
Q69
Q70
Q51
Q52
Q53
Q54
Q63
Q64
Q65
Q66
Q71
Q72
LED_GPIO_31
LED_GPIO_32
LED_GPIO_33
LED_GPIO_34
LED_GPIO_35
LED_GPIO_41
LED_GPIO_42
LED_GPIO_43
LED_GPIO_44
LED_GPIO_45
R94 10K
R95 10K
R96 10K
R97 10K
R106 10K
R107 10K
R108 10K
R109 10K
R116 10K
R117 10K
LED Array

SAMD21J provides 6 SERCOMs which can be configured as UART's and other Protocols.
Programmer must read Datasheet for further info.

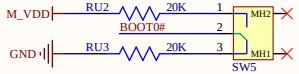


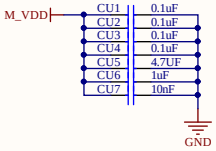
Signals are connected to the GPIOs of SAMD21J except UART's. Programmer must read Datasheet for further info.

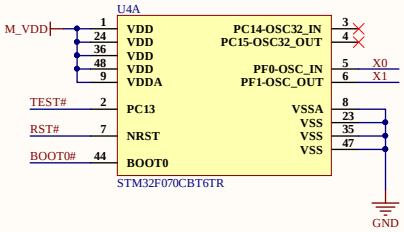


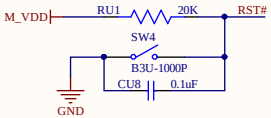


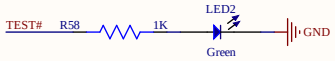




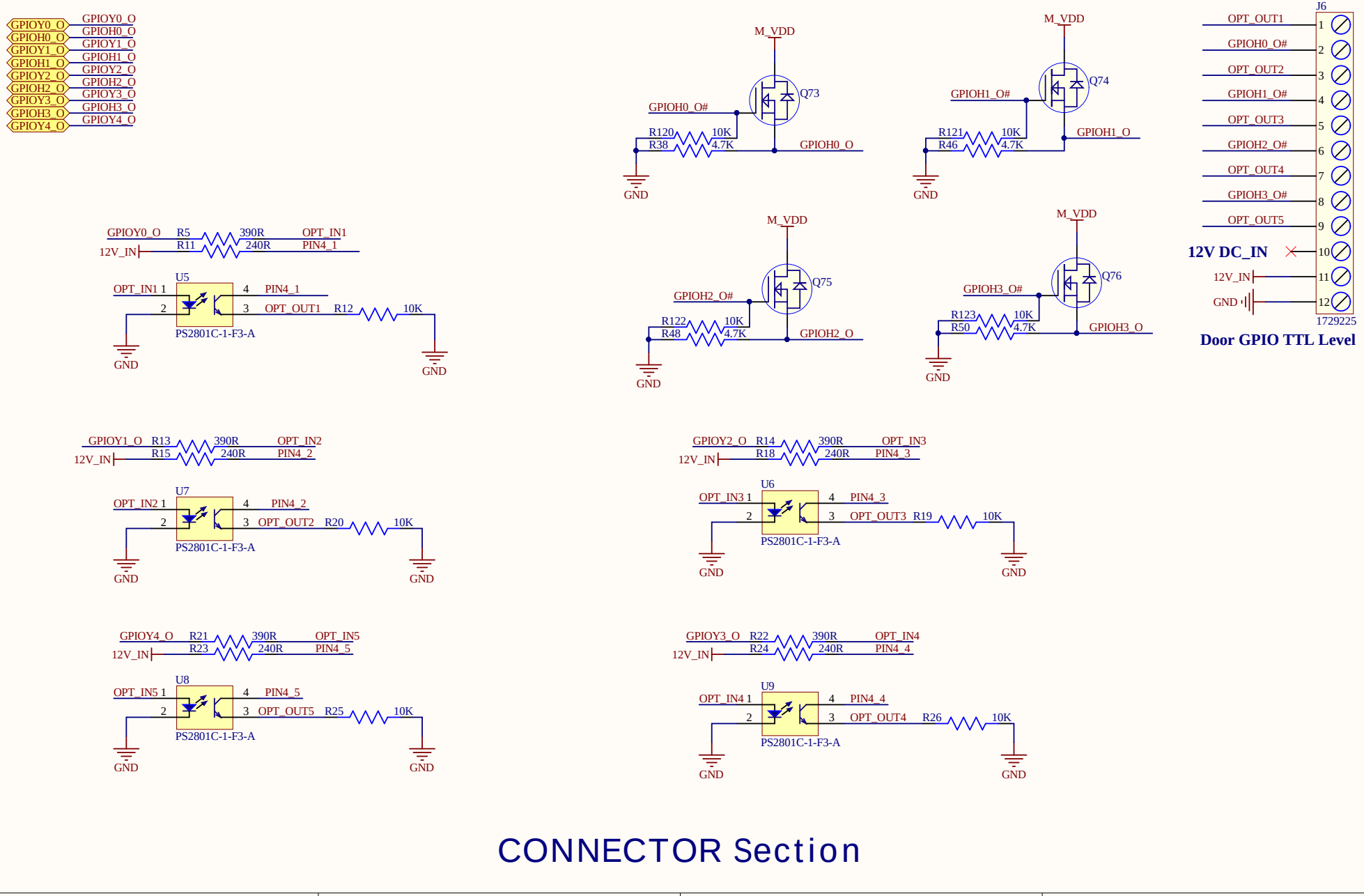








Microcontroller Section


GPIOY0_O
GPIOH0_O
GPIOY1_O
GPIOH1_O
GPIOY2_O
GPIOH2_O
GPIOY3_O
GPIOH3_O
GPIOY4_O
M_VDD
Q73
GPIOH0_O#
R120
10K
R38
4.7K
GPIOH0_O
GND
M_VDD
Q74
GPIOH1_O#
R121
10K
R46
4.7K
GPIOH1_O
GND
J6
OPT_OUT1
GPIOH0_O#
OPT_OUT2
GPIOH1_O#
OPT_OUT3
GPIOH2_O#
OPT_OUT4
GPIOH3_O#
OPT_OUT5
12V DC_IN
12V_IN
GND
1729225
Door GPIO TTL Level
GPIOY0_O
R5
390R
OPT_IN1
12V_IN
R11
240R
PIN4_1
U5
OPT_IN1
PIN4_1
OPT_OUT1
R12
10K
PS2801C-1-F3-A
GND
GND
M_VDD
Q75
GPIOH2_O#
R122
10K
R48
4.7K
GPIOH2_O
GND
M_VDD
Q76
GPIOH3_O#
R123
10K
R50
4.7K
GPIOH3_O
GND
GPIOY1_O
R13
390R
OPT_IN2
12V_IN
R15
240R
PIN4_2
U7
OPT_IN2
PIN4_2
OPT_OUT2
R20
10K
PS2801C-1-F3-A
GND
GND
GPIOY2_O
R14
390R
OPT_IN3
12V_IN
R18
240R
PIN4_3
U6
OPT_IN3
PIN4_3
OPT_OUT3
R19
10K
PS2801C-1-F3-A
GND
GND
GPIOY4_O
R21
390R
OPT_IN5
12V_IN
R23
240R
PIN4_5
U8
OPT_IN5
PIN4_5
OPT_OUT5
R25
10K
PS2801C-1-F3-A
GND
GND
GPIOY3_O
R22
390R
OPT_IN4
12V_IN
R24
240R
PIN4_4
U9
OPT_IN4
PIN4_4
OPT_OUT4
R26
10K
PS2801C-1-F3-A
GND
GND
CONNECTOR Section